

Mobile app development is in a continuous state of change. A new operating system (OS) typically rolls out every six months or every year, with a few unique features that mobile apps can utilise. Developing mobile apps for a specific version of the operating system (OS) or even developing them for a native OS is much like the story of Goldilocks — requiring that multiple solutions be tried out before finding the perfect fit for one's development needs.

Mobile app development platforms

Mobile application development platforms, also referred to as mobile enterprise application platforms, belong to the group of both products and services that help mobile apps to be developed in a systematic order. With the presence of a wide variety of mobile devices, networks and user groups, the development of mobile software applications is continuously becoming more challenging and difficult. Using a mobile app development platform can help in handling and overcoming this challenge. Such platforms allow a business to rapidly build a mobile app, test it and then deploy it on different smartphones or tablets. The platforms can manage different devices, both when they are deployed or even during the entire life cycle of a mobile app.

Gartner has been credited with coining the term 'mobile application development platform'. These platforms consist of both a mobile client application as well as a mobile middleware server. The latter does not store any data sets in itself, but it manages their security, scalability, systems integration, cross-platform support, communications and more. The client applications then connect with the specified server, and hence are the driving force behind both the applied business logic and the user interface on any given mobile device.

A business can have its own mobile application development platform built or it can buy one of the many third party products available. These platforms include various features, the most significant of which are:

1. They are equipped with front-end development tools and management tools for application programming interfaces (APIs).
2. They also provide the mobile Backend as a Service (BaaS) feature.
3. They possess native, Web and hybrid mobile app development capabilities.
4. They include mobile application management (MAM) tools that are used for deploying and securing the apps.
5. These platforms implement both inclusive and long-term approaches, which are really helpful for standalone mobile apps.
6. The mobile software application is developed once and then deployed in all the different mobile devices including tablets, smartphones or notebooks.
7. They ensure that the mobile app is compatible with each mobile device that it is sent to, without changing the way the application works.
8. They help ensure adherence to the Rule of Three, which



Figure 1: The different types of mobile apps developed (source: *googleimages.com*)

refers to the need for mobile solutions that work together with three or more back-end data sources, are compatible with three or more mobile applications or work with three or more operating systems.

Criteria for choosing a mobile application development platform

Going cross-platform, staying native or going cross-platform while staying native – these are some of the choices available to developers. Such options can create quite a bit of confusion for someone who simply wants to develop a mobile app. So here are some of the factors, which if considered, can clear this confusion to a great extent.

1. **Different types of devices supported:** The very first step in choosing the right solution is to determine the types of devices each platform supports. A list of tools that enable us to build native applications, cross-platform applications or both for popular mobile operating systems (iOS, Android and Windows) and different devices (tablets and smartphones) should be created. This differentiation becomes easier when a platform supports additional operating systems and devices.
2. **Integration with third party systems:** An effective mobile application development platform should have the ability to seamlessly integrate with different third party technologies and tools that impact the app throughout its life cycle. The platform must not only ensure seamless integration with other services and systems but also allow developers to create applications that support mobile app management. This integration should extend to different existing back-end systems and the continuous delivery infrastructure.
3. **Integrated development environment (IDEs):** When we analyse various mobile application development platforms, we should consider all the options available. The future requirements of the app, considering that they are likely to change, should be planned for. Platforms differ in the development features they provide. Some platforms provide modern IDEs, which can be installed on the machines of developers, while other platforms provide Web based plugins